CIS 5810 FINAL PROJECT · GROUP [#]

Magic Eraser

Type what to remove from a video. It disappears.



Input



“remove the person”

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github.com/Norrozo/5810finalproject

THE PROBLEM

Removing an object from video takes hours of frame-by-frame editing

What makes it hard

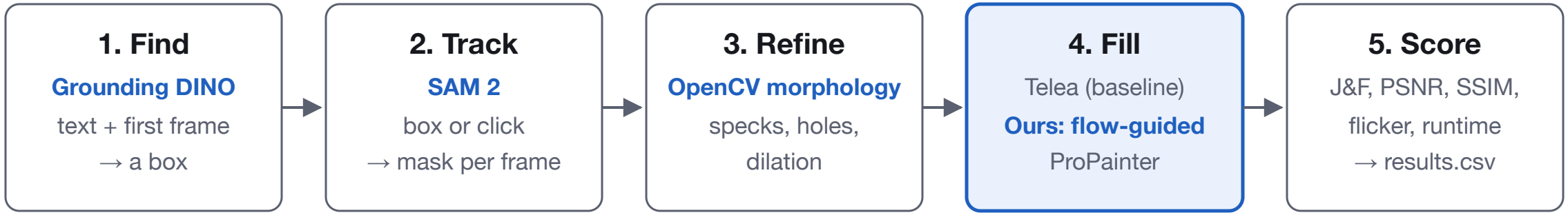
- **Find** the object from a plain description
- **Track** it through every frame as it moves and turns
- **Fill** what was behind it, which the camera may never have seen
- Do it **consistently**, so the fill doesn't shimmer frame to frame

Our goal

- Input: a 5–10 s clip + text (“remove the dog”)
- Output: the same clip, object gone, no blur or flicker
- Audience: casual creators removing a photobomber without After Effects
- Measure every change against a fixed baseline

SYSTEM DESIGN

Five stages, each a separate script that reads and writes files



Swap one stage, keep the rest fixed → every iteration changes exactly one thing.

Why files between stages: a crashed run resumes where it stopped, and later versions reuse the baseline's masks, so fill methods are compared on identical masks.

Same code everywhere: the benchmark, the Colab notebook and the web demo all call these five stages.

SAM 2: Meta 2024 · Grounding DINO: IDEA 2023 · ProPainter: ICCV 2023

HOW WE MEASURE

Real videos have no answer key, so we build one



Input: a DAVIS dog pasted onto another video



Tracking truth: the exact paste mask



Fill truth: the untouched video

- **Test set:** 10 clips from DAVIS *val*, frozen before any results
- **Tuning set:** 5 clips from DAVIS *train*. Settings are chosen here, never on the test set

- **J&F** ↑ did we track the right pixels?
- **PSNR mask** ↑ is the filled hole close to the truth? (dB)
- **Flicker** ↓ does the fill shimmer? (clean video: 7.6)

DAVIS 2017, 480p, 40–60 frames per clip

BASELINE (B0): CLICK → SAM 2 → OPENCV TELEA

Tracking is nearly perfect; the fill is a fog



Input



B0 result

Telea fills the hole by smudging the colors at its edge inward. It never looks at other frames, so a big hole becomes a smear, and the smear changes every frame.

0.943

J&F: tracking is nearly solved

16.4 dB

PSNR inside the hole: visibly wrong

2.5×

the clean video's flicker

Means over 10 test clips · 1.38 s/frame, 97% of it SAM 2

ITERATIONS I1 AND I2

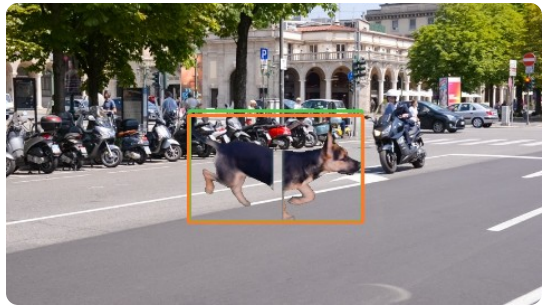
Bigger masks hurt; text prompts work 9 of 10 times

I1 Grow the mask 5 px

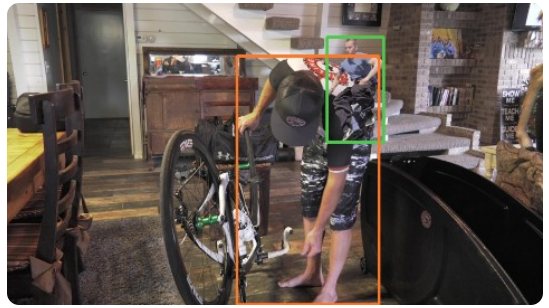
PSNR mask 16.39 → **15.51**. Pasted objects have razor-sharp edges, so there's no fuzzy halo to clean up. A bigger mask is just a bigger hole to blur.

A negative result that exposes a limit of our benchmark. On a real video (a bear's fur) a halo does appear, so we'll re-test on real clips.

I2 Type the object's name



“dog”: one box covers both halves. J&F 0.60 → 0.95



“person”: picked the wrong person. J&F 0.11

■ true object ■ Grounding DINO's box

Grounding hit = box IoU > 0.5 with the true object

ITERATION I3: OUR OWN METHOD

Flow-guided fill: copy the real background from other frames

1. **Optical flow** (Farnebäck): where did each pixel move between neighboring frames?
2. **Complete the flow**: inside the hole the flow follows the dog, so replace it with the smooth background motion around it.
3. **Propagate**: follow the flow to a neighbor frame; if the background is visible there, copy it. Sweep forward and backward.
4. **Fallback**: whatever no frame ever saw goes to Telea.



B0: fog

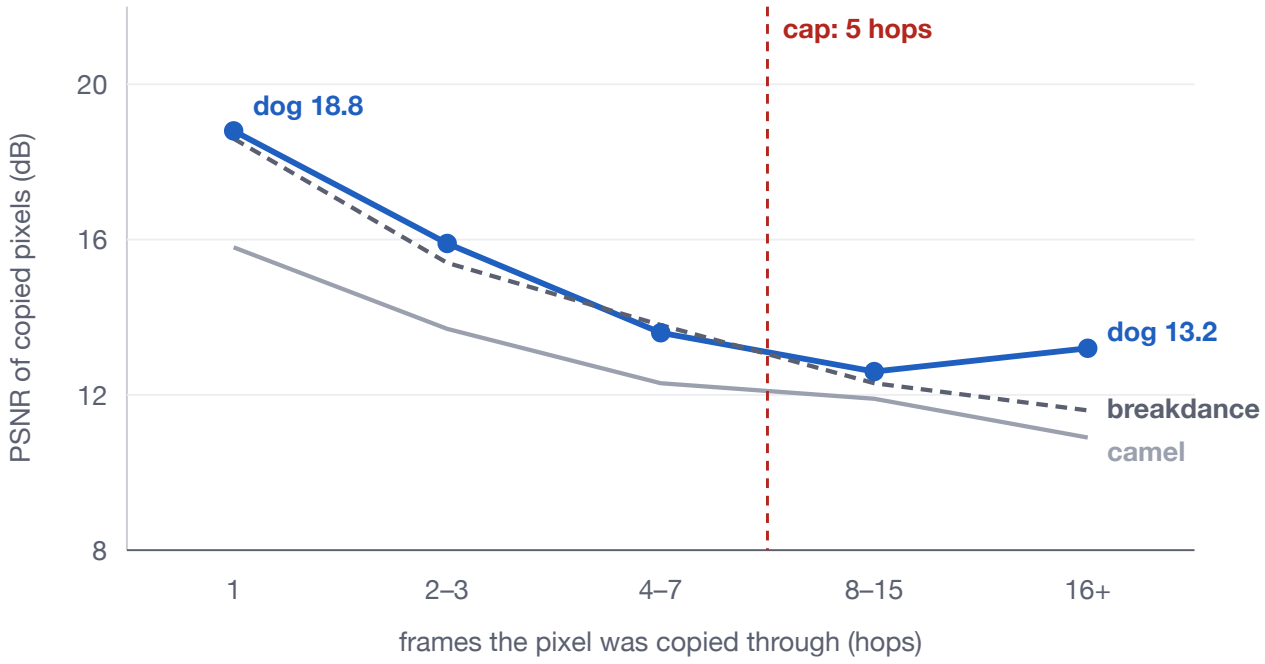


Ours: real cart recovered

Flicker **–13%**, because the same real background is copied into every frame. But PSNR *dropped* at first (16.39 → 15.98). Why?

ITERATION I3B: DEBUGGING OUR METHOD

Each extra hop costs accuracy, so we cap hops at 5



Each hop adds a little optical-flow error and a little interpolation blur. After 8+ hops, copied pixels are **worse than Telea**.

We chose the cap on the **tuning set** (5 → 20.65 dB vs 19.46 uncapped), then ran it once on the test set.

	PSNR mask	Flicker
B0	16.39	18.75
I3b (ours)	**16.50**	**16.31**

Hop analysis on B0 masks; chart shows 3 test clips

ITERATION I4: PROPAINTER

A learned video inpainter removes the fog



Input | B0 Telea | I3b ours | **I4 ProPainter** | Truth

Same idea as ours, every step learned: RAFT flow, a network that completes flow, propagation in image and feature space, and a **transformer that invents** background no frame ever showed. That last step is exactly what our method lacks.

+4.75
dB PSNR mask vs B0

–51%
flicker vs B0

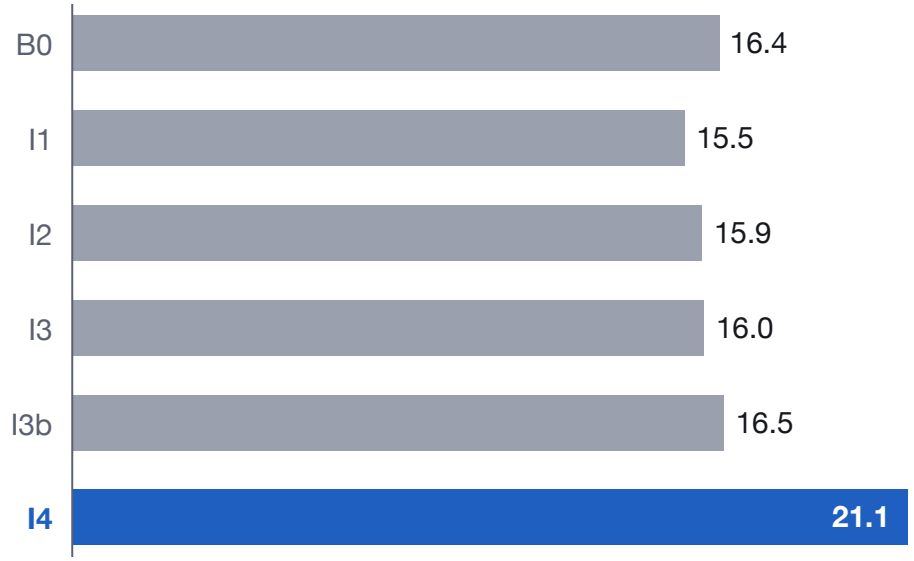
Better on all 10 clips. Cost: about 4× runtime.

Zhou et al., ProPainter, ICCV 2023 · run in its own Python environment

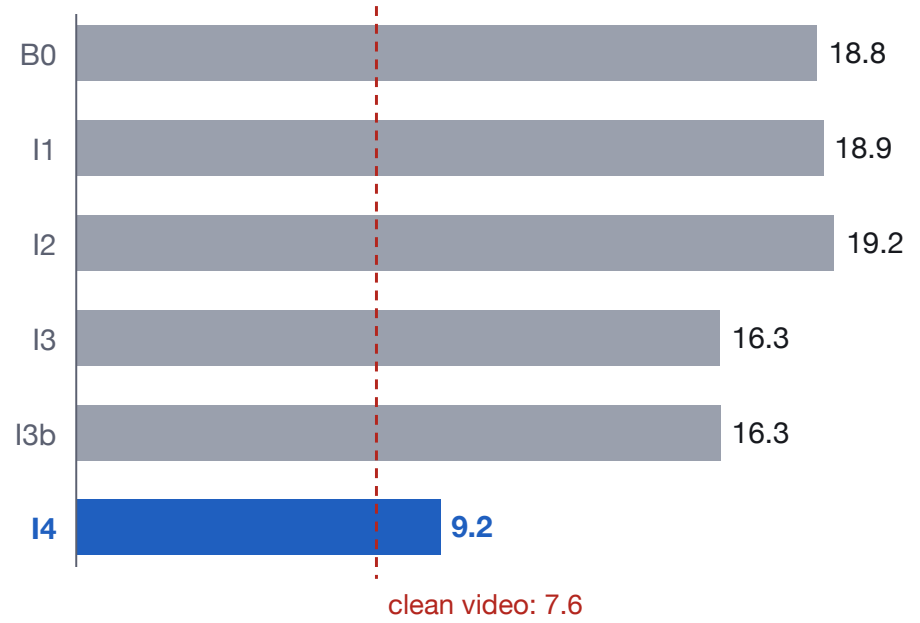
RESULTS SO FAR

ProPainter cut the error and halved the flicker

PSNR inside the hole (dB) ↑



Flicker ↓



Means over the 10 test clips. Tracking (J&F 0.943) is identical for all fill methods because they share the baseline's masks; I2 is 0.890 due to the one grounding miss.

REAL VIDEOS: TEXT PROMPT → SAM 2 → PROPAINTER

On real clips it works when the background moves past the object



“remove the ball”



clean



“remove the person”



clean, faint shadow



“remove the tennis player”



racket left floating



“remove the bear”



smudge: camera follows the bear

DAVIS clips not used in the benchmark

The remaining failures are about *what* to remove, not how to fill

Lessons

- Measure before fixing: the hop analysis turned a failed iteration into a fix
- Tune on a separate set, or the test numbers lie
- PSNR rewards blur, so always look at the videos too
- A pasted benchmark hides real effects: soft edges, shadows

Next iterations

When	Iteration
Oct	LPIPS metric; re-test I1 on real clips
Nov 20	**I5** re-detect every N frames · **I6** speed (< 1 s/frame)
Dec 6	**I7** multiple objects (“the player and the racket”), shadows

LIVE DEMO

Upload a video. Type what to remove.

Gradio app on the same five stages · `python -m src.demo` · Colab: `notebooks/colab.ipynb`



Input



“remove the person”